Lab 09

Layer 2 Security and Layer 2 VLAN Security

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9.1 Layer 2 Security

9.1.1 Objective

The Objectives of this lab are:

- Assign the Central switch as the root bridge.
- Secure spanning-tree parameters to prevent STP manipulation attacks.
- Enable port security to prevent CAM table overflow attacks.

9.1.2 Background/Scenario

There have been a number of attacks on the network recently. For this reason, the network administrator has assigned you the task of configuring Layer 2 security.

For optimum performance and security, the administrator would like to ensure that the root bridge is the 3560 Central switch. To prevent spanning-tree manipulation attacks, the administrator wants to ensure that the STP parameters are secure. To prevent against CAM table overflow attacks, the network administrator has decided to configure port security to limit the number of MAC addresses each switch port can learn. If the number of MAC addresses exceeds the set limit, the administrator would like the port to be shutdown.

All switch devices have been preconfigured with the following

- Console password: **ciscoconpa55**
- Enable password: **ciscoenpa55**
- SSH username and password: **SSHadmin/ ciscosshpa55**

9.1.3 Topology

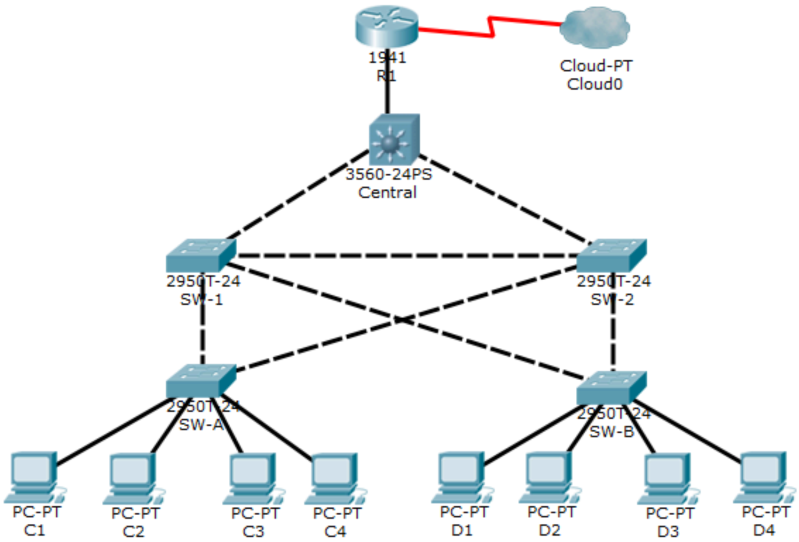


Figure 1: Topology

Task 1.1: Configure Root Bridge

1. Determine the current root bridge

   From Central, issue the **show spanning-tree** command to determine the current root bridge, to see the ports in use, and to see their status.

```
Central#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0009.7C61.9058
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.D31C.634C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Root FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p
```

   Which switch is the current root bridge?

   SW1 is the root bridge

2. Assign Central as the primary root bridge.

Using the **spanning-tree vlan 1 root primary** command and assign **Central** as the root bridge.

```
Central#
Central#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Central(config)#spanning-tree vlan 1 root primary
Central(config)#end
Central#
%SYS-5-CONFIG_I: Configured from console by console

Central#write memory
Building configuration...
[OK]
```

3. Assign SW-1 as a secondary root bridge.

   Assign **SW-1** as the secondary root bridge using the **spanning-tree vlan 1 root secondary** command.

```
SW-1>en
Password:
SW-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-1(config)#spanning-tree vlan 1 root secondary
SW-1(config)#end
SW-1#
%SYS-5-CONFIG_I: Configured from console by console

SW-1#write memory
Building configuration...
[OK]
```

4. Verify the spanning-tree configuration

   Issue the **show spanning-tree** command to verify that **Central** is the root bridge.

```
Central#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00D0.D31C.634C
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     00D0.D31C.634C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Gi0/1            Desg FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p

Central#
```

Which switch is the current root bridge?

Central switch

Task 1.2: Protect Against STP Attacks

Secure the STP parameters to prevent STP manipulation attacks

1. Enable PortFast on all access ports.

   PortFast is configured on access ports that connect to a single workstation or server to enable them to become active more quickly. On the connected access ports of the **SW-A** and **SW-B,** use the **spanning-tree portfast** command.

```
SW-A>en
Password:
SW-A#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-A(config)#interface range f0/1 - 4
SW-A(config-if-range)#spanning-tree portfast
```

2. Enable BPDU guard on all access ports.

   BPDU guard is a feature that can help prevent rogue switches and spoofing on access ports. Enable BPDU guard on **SW-A** and **SW-B** access ports.

```
SW-A(config-if-range)#spanning-tree bpduguard enable
SW-A(config-if-range)#exit
```

Note: Spanning-tree BPDU guard can be enabled on each individual port using the **spanning-tree bpduguard enable** command in interface configuration mode or the **spanning-tree portfast bpduguard default** command in global configuration mode. For grading purposes in this activity, please use the **spanning-tree bpduguard enable** command.

3. Enable root guard

```
Enter configuration commands, one per line.  End
SW-1(config)#interface range f0/23 - 24
SW-1(config-if-range)#spanning-tree guard root
SW-1(config-if-range)#exit
SW-1(config)#
```

```
SW-2>en
Password:
SW-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-2(config)#interface range f0/23 - 24
SW-2(config-if-range)#spanning-tree guard root
SW-2(config-if-range)#exit
```

Root guard can be enabled on all ports on a switch that are not root ports. It is best deployed on ports that connect to other non-root switches. Use the **show spanning-tree** command to determine the location of the root port on each switch.

On **SW-1**, enable root guard on ports F0/23 and F0/24. On **SW-2**, enable root guard on ports F0/23 and F0/24.

Task 1.3: Configure Port Security and Disable Unused Ports

1. Configure basic port security on all ports connected to host devices.

This procedure should be performed on all access ports on **SW-A** and **SW-B**. Set the maximum number of learned MAC addresses to **2**, allow the MAC address to be learned dynamically, and set the violation to **shutdown**.

Note: A switchport must be configured as an access port to enable port security.

```
SW-A(config)#
SW-A(config)#interface range f0/1 - 4
SW-A(config-if-range)#switchport mode access
SW-A(config-if-range)#switchport port-security
SW-A(config-if-range)#switchport port-security maximum 2
SW-A(config-if-range)#switchport port-security mac-address sticky
SW-A(config-if-range)#switchport port-security violation shutdown
SW-A(config-if-range)#exit
```

Why is port security not enabled on peorts that are connected to other switch devices?

Trunk links carry traffic from many devices. Applying port security with a low MAC address limit would cause the port to shut down due to legitimate traffic, which severs network connectivity.

2. Verify port security.

a. On **SW-A**, issue the command **show port-security interface f0/1** to verify that port security has been configured.

SW-A# **show port-security interface f0/1**

```
SW-A#show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

b. Ping from C1 to C2 and issue the command **show port-security interface f0/1** again to verify that the switch has learned the MAC address for **C1**.

```
Pinging 10.1.1.11 with 32 bytes of data:

Reply from 10.1.1.11: bytes=32 time=76ms TTL=128
Reply from 10.1.1.11: bytes=32 time=1ms TTL=128
Request timed out.
Reply from 10.1.1.11: bytes=32 time=44ms TTL=128

Ping statistics for 10.1.1.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 76ms, Average = 40ms
```

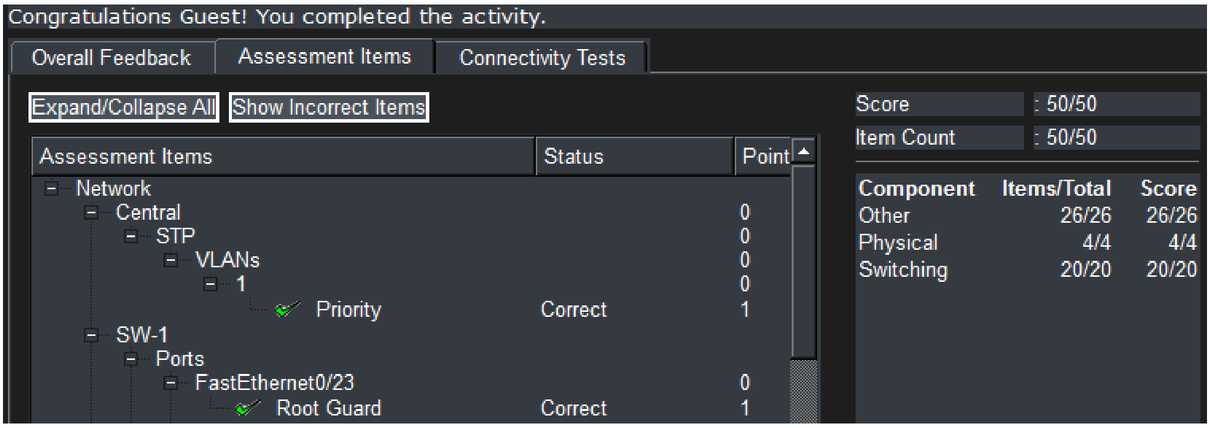
```
SW-A#show port-security interface f0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0060.3E81.4647:1
Security Violation Count   : 0
```

3. Disable unused ports.

   Disable all ports that are currently unused.

```
SW-A(config)#interface range f0/5 - 22
SW-A(config-if-range)#shutdown

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down

%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
```

4. Check Results.



9.2 Layer 2 VLAN Security

8.2.1 Objectives:

- Connect a new redundant link between SW-1 and SW-2.
- Enable trunking and configure security on the new trunk link between SW-1 and SW-2.
- Create a new management VLAN (VLAN 20) and attach a management PC to that VLAN.
- mplement an ACL to prevent outside users from accessing the management VLAN.

9.2.2 Background/Scenario:

A company's network is currently set up using two separate VLANs: VLAN 5 and VLAN 10. In addition, all trunk ports are configured with native VLAN 15. A network administrator wants to add a redundant link between switch SW-1 and SW-2. The link must have trunking enabled and all security requirements should be in place.

In addition, the network administrator wants to connect a management PC to switch SW-A. The administrator would like to enable the management PC to connect to all switches and the router but does not want any other devices to connect to the management PC or the switches. The administrator would like to create a new VLAN 20 for management purposes.

All devices have been preconfigured with:

- Console password: **ciscoconpa55**
- Enable password: **ciscoenpa55**
- SSH username and password: **SSHadmin/ ciscosshpa55**

8.2.3 Topology:

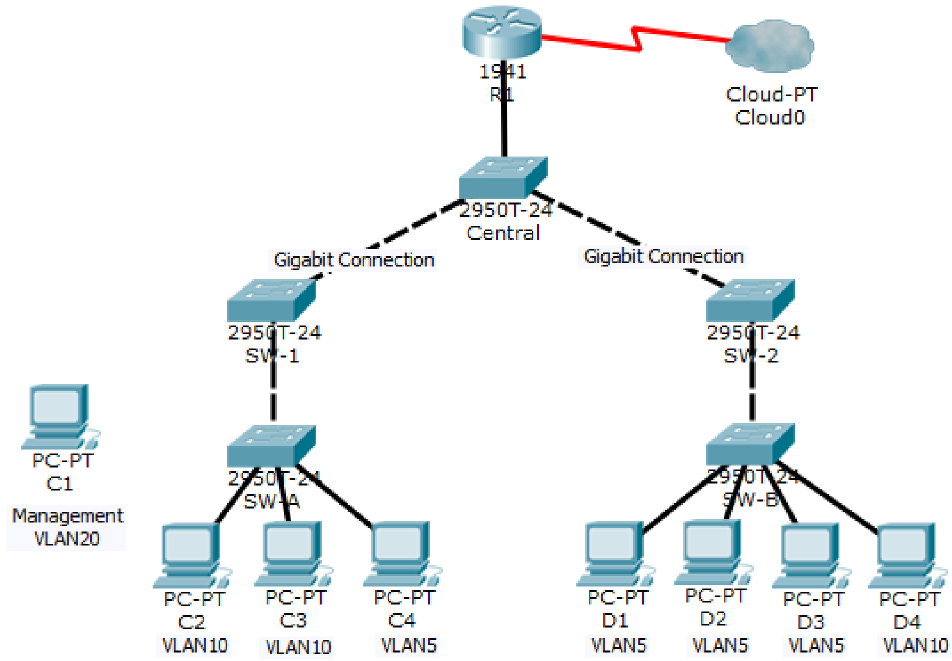


Figure 2

Task 2.1: Verify Connectivity

1. Verify connectivity between C2 (VLAN 10) and C3 (VLAN 10).

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

2. Verify connectivity between C2 (VLAN 10) and D1 (VLAN 5)

```
C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Reply from 192.168.5.2: bytes=32 time=10ms TTL=127
Reply from 192.168.5.2: bytes=32 time<1ms TTL=127
Reply from 192.168.5.2: bytes=32 time<1ms TTL=127
Reply from 192.168.5.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
```

Note: If using the simple PDU GUI Packet, be sure to ping twice to allow for ARP

Task 2.2: Create a Redundant Link Between SW-1 and SW-2

1. Connect SW-1 and SW-2.

   Using a crossover cable, connect port F0/23 on **SW-1** to port F0/23 on **SW-2**.

2. Enable trunking, including all trunk security mechanisms on the link between SW-1 and SW-2.

   Trunking has already been configured on all pre-existing trunk interfaces. The new link must be configured for trunking, including all trunk security mechanisms. On both **SW-1** and **SW-2**, set the port to trunk, assign native VLAN 15 to the trunk port, and disable auto-negotiation.

```
Enter configuration commands, one per line.  End
SW-1(config)#int f0/23
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#switchport trunk native vlan 15
SW-1(config-if)#switchport nonegotiate
SW-1(config-if)#
```

```
SW-2(config)#int f0/23
SW-2(config-if)#switchport mode trunk
SW-2(config-if)#switchport trunk native vlan 15
SW-2(config-if)#switchport nonegotiate
```

Task 2.3: Enable VLAN 20 as a Management VLAN

The network administrator wants to access all switch and routing devices using a management PC. For security purposes, the administrator wants to ensure that all managed devices are on a separate VLAN.

1. Enable a management VLAN (VLAN 20) on SW-A.

a. Enable VLAN 20 on **SW-A.**

```
SW-A(config)#vlan 20
SW-A(config-vlan)#name management
SW-A(config-vlan)#exit
```

b. Create an interface VLAN 20 and assign an IP address within the 192.168.20.0/24 network

```
SW-A(config)#int vlan 20
SW-A(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW-A(config-if)#ip address 192.168.20.11 255.255.255.0
SW-A(config-if)#no shutdown
```

2. Enable the same management VLAN on all other switches.

a. Create the management VLAN on all switches: **SW-B, SW-1, SW-2**, and **Central**

```
SW-B(config)#vlan 20
SW-B(config-vlan)#name Management
SW-B(config-vlan)#exit
```

```
SW-1(config)#vlan 20
SW-1(config-vlan)#name Management
SW-1(config-vlan)#exit
```

```
SW-2(config)#vlan 20
SW-2(config-vlan)#name Management
SW-2(config-vlan)#exit
```

```
Central(config)#vlan 20
Central(config-vlan)#name Management
Central(config-vlan)#exit
```

```
SW-A#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Gig0/1
                                                Gig0/2
5    VLAN0005                         active    Fa0/4
10   VLAN0010                         active    Fa0/2, Fa0/3
15   VLAN0015                         active
20   Management                       active    Fa0/1
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

b. Create an interface VLAN 20 on all switches and assign an IP address within the 192.168.20.0/24 network.

```
SW-B(config)#int vlan 20
SW-B(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW-B(config-if)#ip address 192.168.20.12 255.255.255.0
SW-B(config-if)#no shutdown
```

```
SW-1(config)#int vlan 20
SW-1(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW-1(config-if)#ip address 192.168.20.13 255.255.255.0
SW-1(config-if)#no shutdown
```

```
SW-2(config)#int vlan 20
SW-2(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW-2(config-if)#ip address 192.168.20.14 255.255.255.0
SW-2(config-if)#no shutdown
```

```
Central(config)#int vlan 20
Central(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Central(config-if)#ip address 192.168.20.15 255.255.255.0
Central(config-if)#no shutdown
```

3. Connect and configure the management PC

   Connect the management PC to **SW-A** port F0/1 and ensure that it is assigned an available IP address within the 192.168.20.0/24 network.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
DHCP	Static (selected)
IPv4 Address	192.168.20.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

4. On SW-A, ensure the management PC is part of VLAN 20.

   Interface F0/1 must be part of VLAN 20.

```
SW-A(config)#int f0/1
SW-A(config-if)#switchport access vlan 20
SW-A(config-if)#
```

5. Verify connectivity of the management PC to all switches.

   The management PC should be able to ping **SW-A, SW-B, SW-1, SW-2**, and **Central**

Task 2.4: Enable VLAN 20 as a Management VLAN

1. Enable a new subinterface on router R1
a. Create subinterface g0/0.3 and set encapsulation to dot1q 20 to account for VLAN 20.

```
R1(config)#int g0/0.3
R1(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.3, changed state to up

R1(config-subif)#enca
R1(config-subif)#encapsulation dot
R1(config-subif)#encapsulation dot1Q 20
```

b. Assign an IP address within the 192.168.20.0/24 network.

```
R1(config-subif)#ip address 192.168.20.1 255.255.255.0
R1(config-subif)#no shutdown
R1(config-subif)#exit
```

2. Verify connectivity between the management PC and R1

   Be sure to configure the default gateway on the management PC to allow for connectivity.

3. Enable security

   While the management PC must be able to access the router, no other PC should be able to access the management VLAN.

a. Create an ACL that allows only the Management PC to access the router.

```
R1(config)#access-list 10 permit 192.168.20.100
R1(config)#access-list 10 deny any
```

b. Apply the ACL to the proper interface(s)

```
R1(config)#line vty 0 15
R1(config-line)#access-class 10 in
R1(config-line)#exit
R1(config)#access-list 100 deny ip any 192.169.20.0 0.0.0.255
R1(config)#access-list 100 permit ip any any
R1(config)#int g0/0.3
R1(config-subif)#ip access-group 100 out
R1(config-subif)#exit
```

Note: There are multiple ways in which an ACL can be created to accomplish the necessary security. For this reason, grading on this portion of the activity is based on the correct connectivity requirements. The management PC must be able to connect to all switches and the router. All other PCs should not be able to connect to any devices within the management VLAN.

4. Verify security.

a. Verify only the Management PC can access the router. Use SSH to access R1 with username SSHadmin and password ciscosshpa55.

`PC>` **ssh -l SSHadmin 192.168.20.100**

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ssh -l SSHadmin 192.168.20.1

Password:


R1#
```

b. From the management PC, ping SW-A, SW-B, and R1. Were the pings successful? Explain.

```
C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.11: bytes=32 time<1ms TTL=255
Reply from 192.168.20.11: bytes=32 time<1ms TTL=255
Reply from 192.168.20.11: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

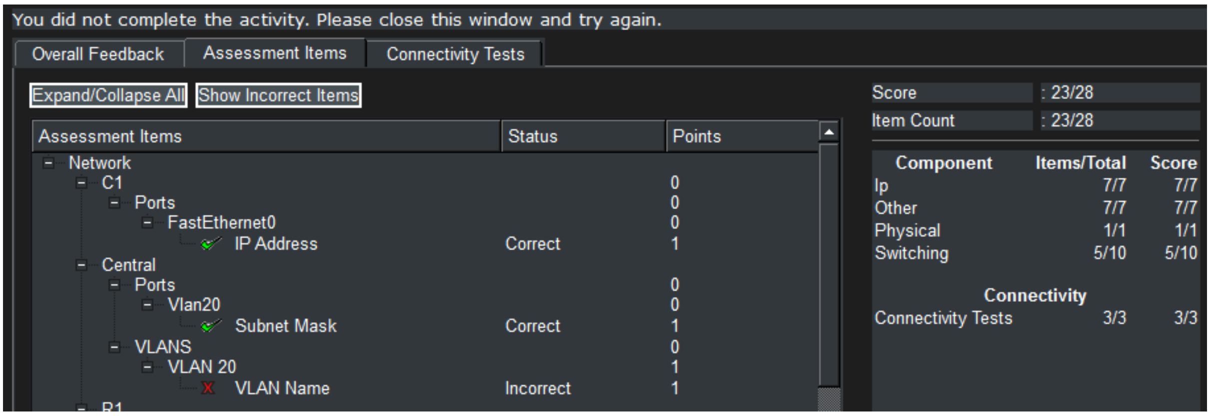
The Management PC, switch interfaces, and router subinterface are on the same subnet and VLAN. Traffic between them is local at Layer 2 and doesn't pass through the router's ACL, which only filters external traffic.

c. From D1, ping the management PC. Were the pings successful? Explain

```
C:\>ping 192.168.20.100

Pinging 192.168.20.100 with 32 bytes of data:

Reply from 192.168.5.100: Destination host unreachable.
Reply from 192.168.5.100: Destination host unreachable.
Reply from 192.168.5.100: Destination host unreachable.
Reply from 192.168.5.100: Destination host unreachable.

Ping statistics for 192.168.20.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

D1 (VLAN 5) and the Management PC (VLAN 20) are on different subnets, so traffic must pass through Router R1. The ping fails because the ACL on R1 blocks all external traffic from entering the management network, causing the router to drop D1's packets.

5. Check results



Connectivity test passes. Error for Vlan name due to case sensitivity.

Assessment Rubric

Lab 09

Layer 2 Security and Layer 2 VLAN Security

Name:	**Student ID:**

Points Distribution

Task No.	**LR 2 Simulation**	**LR5 Results/Plots**	**LR9 Report**
Task 1.1	5	5	
Task 1.2	10	-	
Task 1.3	10	5	
Task 2.1	5	-	
Task 2.2	10	5	
Task 2.3	10	-	
Task 2.4	10	5	
Total	/60	/20	/10
CLO Mapped	CLO 2	CLO 2	CLO2

Affective Domain Rubric		**Points**	**CLO Mapped**
AR 7	Report Submission	/10	CLO 2

CLO	**Total Points**	**Points Obtained**
2	100	
Total	**100**	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.

Lab Evaluation Assessment Rubric

#	Assessment Elements	Level 1: Unsatisfactory Points 0-1	Level 2: Developing Points 2	Level 3:Good Points 3	Level 4:Exemplary Points 4
LR2	**Program/Code / Simulation Model/ Network Model**	Program/code/simulation model/network model does not implement the required functionality and has several errors. The student is not able to utilize even the basic tools of the software.	Program/code/simulation model/network model has some errors and does not produce completely accurate results. Student has limited command on the basic tools of the software.	Program/code/simulation model/network model gives correct output but not efficiently implemented or implemented by computationally complex routine.	Program/code/simulation /network model is efficiently implemented and gives correct output. Student has full command on the basic tools of the software.
LR5	**Results & Plots**	Figures/ graphs / tables are not developed or are poorly constructed with erroneous results. Titles, captions, units are not mentioned. Data is presented in an obscure manner.	Figures, graphs and tables are drawn but contain errors. Titles, captions, units are not accurate. Data presentation is not too clear.	All figures, graphs, tables are correctly drawn but contain minor errors or some of the details are missing.	Figures / graphs / tables are correctly drawn and appropriate titles/captions and proper units are mentioned. Data presentation is systematic.
LR9	**Report**	All the in-lab tasks are not included in report and / or the report is submitted too late.	Most of the tasks are included in report but are not well explained. All the necessary figures / plots are not included. Report is submitted after due date.	Good summary of most the in-lab tasks is included in report. The work is supported by figures and plots with explanations. The report is submitted timely.	Detailed summary of the in-lab tasks is provided. All tasks are included and explained well. Data is presented clearly including all the necessary figures, plots and tables.